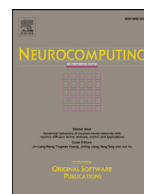




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## Bin loss for hard exudates segmentation in fundus images

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## ABSTRACT

Diabetic retinopathy is one of the leading reasons that causes blindness. And the segmentation of hard exudates in color fundus images is crucial for early diagnosis of diabetic retinopathy, which is a difficult task due to its uncertainty in size, shape and contrast. Class-balanced cross entropy (CBCE) loss is the most popular objective function for image segmentation task to solve the class-unbalance problem. However, we show that background pixels tend to be misclassified to hard exudates in CBCE since the loss for a misclassified background pixels is much smaller than that for a misclassified hard exudate pixel, which is called loss-unbalance problem here. A top-k loss is proposed in this paper, which considers the cases of both class-unbalance and loss-unbalance by focusing more over the hard-to-classify pixels. Moreover, a fast version of the top-k loss, named bin loss, is implemented for efficiency, which reduces the time complexity from  $O(n \log n)$  of top-k loss to  $O(n)$ , where  $n$  is the number of background pixels. We evaluated the proposed bin loss over two public datasets for hard exudates segmentation task, including e-ophtha EX and IDRiD. Furthermore, three popular models for image segmentation, HED, DeepLab v2, and FCNR, were used to evaluate the versatility of bin loss. Extensive experiments show that each model with the proposed bin loss performs better than that with CBCE loss, which demonstrates bin loss is versatile so that it can be applied to different models for performance improvement. Specially, for DeepLab over e-ophtha EX, the F-score increases 5.2 percentage points, and the area under the SE-PPV curve (AUC) increases 10.6 percentage points. Moreover, the AUC increases more than 4 percentage points over IDRiD dataset for both DeepLab and FCNR. The source code of bin loss is available at: [https://github.com/guomugong/bin\\_loss](https://github.com/guomugong/bin_loss).

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## 1. Introduction

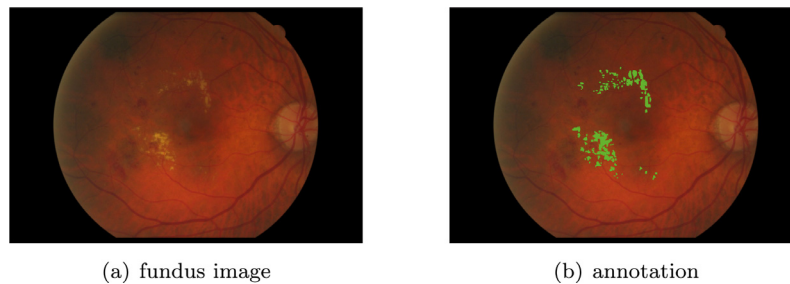
Diabetic Retinopathy (DR) has been one of the leading causes for blindness in working-age people. According to World Diabetes Foundation, there will be over 438 million people with diabetes worldwide by 2030 [1]. Studies show that regular fundus screening and effective interventions in the early stages can significantly reduce the blinding rate of DR. Therefore, it is important to diagnose DR at an earlier clinical stage. By locating small lesions and detecting fundus tissues from the fundus images, patients can be diagnosed as early as possible whether they have DR, and quantitative analysis of the disease can make the patients get the best treatment to reduce the risk of blindness [2].

In clinical practice, exudates, including hard ones and soft ones, are usually the basis for diagnosis of DR, and their size, number

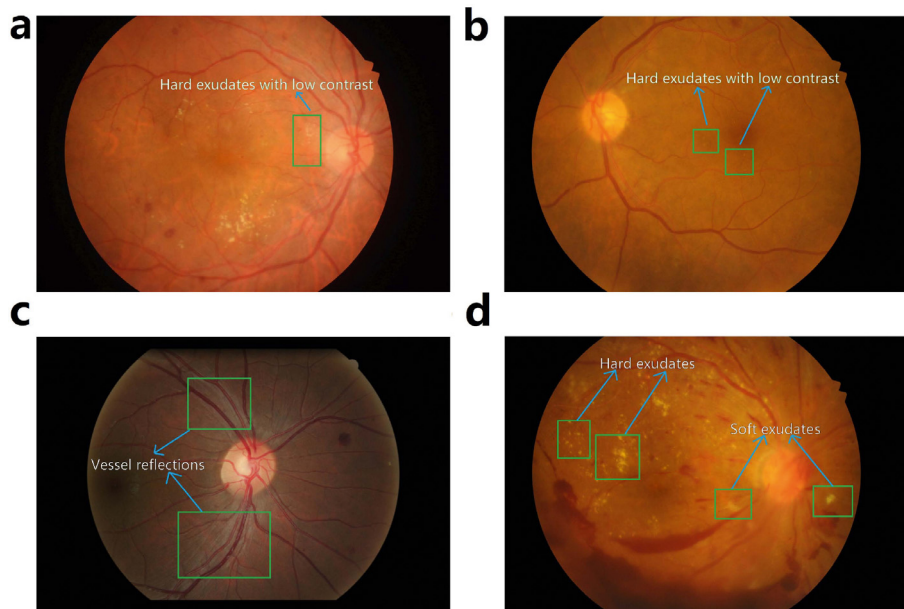
and locations can be used to evaluate the severity level of DR. As is shown in Fig. 1, hard exudates appear as bright yellow crystalline granules having sharper borders. Since hard exudates are crucial indicators for the early DR, it is important to accurately detect them for timely treatment. However, analysis and detection of hard exudates by human are time-consuming, and remarkable skills are required. In the region with poor medicine conditions, late diagnosis of DR may cause people suffer high risk of blinding, it being urgent to develop a tool for automatically segmenting hard exudates from the fundus image. Many efforts have been made in recent studies [3], and the performance for hard exudates segmentation was constantly improved. In addition, low contrast region detection in fundus images are nicely handled in [4,5]. However, it is still a challenging task due to varied sizes, low contrast and structures that is similar with hard exudates, some challenges being shown in Fig. 2.

Benefitting from big data and the improvement of computing capacity, deep learning methods achieved significant performance in many areas, especially in computer vision and medical image

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**Fig. 1.** An example of fundus image in e-optha EX dataset that contains hard exudates. (a) retinal fundus image and (b) annotation provided by the ophthalmologist (hard exudate pixels are marked as green).



**Fig. 2.** Retinal images from e-optha EX and IDRiD datasets. (a) and (b) The contrast of hard exudate pixels is low. (c) and (d) Some structures are highly similar to hard exudates.

analysis [2,6–12]. Recently, deep learning based methods were also applied to hard exudates segmentation [3,13,14]. Hard exudates segmentation is in essence a binary classification problem, which classifies hard exudate pixels as one class and background pixels as another class. Since the distribution of hard exudate pixels and background exudate pixels is highly imbalanced, the sigmoid cross entropy loss function is not appropriate which tends to misclassify hard exudate pixels to background ones. Therefore, researchers used class-balanced cross entropy (CBCE) loss [8] to segment hard exudates [3] for better performance. As a result, each hard exudate pixel is given higher weights, while each background pixel is given lower weights. However, as is shown in Fig. 2, there exists many structures that is similar with hard exudate pixels in retinal fundus images, so that the CBCE loss is tend to misclassify those structures as hard exudates since the loss for a misclassified background pixel is much smaller than that for a misclassified hard exudate pixel, which is called loss-unbalance problem here. In this paper, we propose a top-k cross entropy loss (written as top-k loss for simplification), which considers the cases of both class-unbalance and loss-unbalance by focusing more over hard-to-classify pixels. Moreover, to reduce the time complexity of selecting the top-k loss value, we designed a fast implementation of the top-k loss and we call it bin loss. We evaluated the bin loss on three popular segmentation models, HED [8], DeepLab v2 [15] and FCNR [3], over two public datasets, e-optha EX [16] and IDRiD [17]. Connected component level validation was used to evaluate the performance of hard exudates segmentation. Experimental results show that the

positive predictive value (PPV), F-score and the area under the SE-PPV curve (AUC) are improved on each model over all datasets. Specially, for DeepLab over e-optha EX dataset, the F-score increases 5.2 percentage points, and the AUC increases 10.6 percentage points. Moreover, for HED over e-optha EX dataset, the sensitivity (SE) increases 3.3 percentage points, the PPV increases 7.3 percentage points, the F-score increases 5.2 percentage points, and the AUC increases 4.3 percentage points. Furthermore, for DeepLab and FCNR over IDRiD dataset, the AUC increases more than 4 percentage points. All these statistical results show the superior performance of bin loss over CBCE loss.

The main contributions of this paper are:

- We proposed a top-k loss to avoid both the class-unbalance problem in CE loss and the loss-unbalance problem in CBCE loss.
- We proposed a fast version of the top-k loss, named bin loss, which can reduce the time complexity from  $O(n \log n)$  of the top-k loss to  $O(n)$ .
- We conducted extensive experiments on three deep models over two hard exudates segmentation datasets. Both visual results and statistical results show that each model with bin loss performs better than that with CBCE loss, verifying the versatility of the bin loss.

The remainder of this paper is organized as follows. Section 2 introduce related works about hard exudates segmentation and class imbalance. We describe the bin loss in detail

in Section 3. Experiments and discussion are given in Section 4. Conclusions are drawn in Section 5.

## 2. Related work

### 2.1. Hard exudates segmentation

A number of methods have been proposed in recent years for hard exudates segmentation, which generally were divided into mathematical morphology based methods, region growing based methods, thresholding based methods, and pattern recognition based methods in previous studies [3,13,14,18]. Here we divide the hard exudates segmentation methods into three categories from the aspect of *Machine Learning*, including Un-Machine learning based methods (manually designed features+manually designed classification rules) [19–24], Semi-Machine learning based methods (manually designed features+learnable classification model) [18,25–28], and Full-Machine learning based methods (learnable features+learnable classification model, i.e., deep learning) [3,13,14].

*Un-machine learning based methods:* Both the features used for classifying hard exudate pixels from background pixels and the classification rules are designed by human, machine learning being not applied at any stage. Kaur and Mittal [19] employed multi-scale matched filter to segment blood vasculature map, and they employed circular Hough transform to detect optic disc. Then, vessels and optic disc were removed from retinal image. Finally, a dynamic thresholding was applied to produce the exudates segmentation result. Welfer et al. [20] proposed a coarse to fine exudates segmentation method, in which a few morphology operators were used, such as regional minima detection, morphological reconstruction and H-maxima transform. Imani and Pourreza [21] proposed a novel method for exudates segmentation by using morphological component analysis. Li and Chutape [22] combined region growing and edge detection to segment exudates. Sinthanayothin et al. [23] used a recursive region growing segmentation algorithm for exudate detection. Sánchez et al. [24] used a mixture model to approximate the histogram of the image, and a dynamic threshold was calculated with the statistical information. Moreover, a post-processing procedure was applied to distinguish hard exudates from other similar bright lesions, such as cotton wool spots and vessels reflection. Since both the features and the classification rules are designed by human, many hyper-parameters are needed for Un-Machine learning methods. As a result, the methods are sensitive to the hyper-parameter settings, method with the pre-defined hyper-parameters performing well over one dataset but poor over another dataset. Moreover, when the number of the images used for designing features and rules increases, the workload is also increased for human.

*Semi-machine learning based methods:* The features used for classification are first designed by human, then a classification model is trained with the features over the training set. García et al. [25] used a combination of global and adaptive thresholding method to generate the candidate exudate regions. Then RBF network was employed to distinguish exudate regions from non-exudate regions. Fraz et al. [18] used an ensemble classifier of bootstrapped decision trees for multi-scale localization and segmentation of exudates in retinal fundus images. Then, morphological reconstruction and contextual cues are applied to reduce misclassified exudates. Somkuwar et al. [26] employed Euclidean distance classifier to classify exudates and non-exudates. The input to the classifier is 6-dimensional intensity based features. Niemeijer et al. [27] used a pixel classifier to identify exudates, drusen and cotton-wool spots. Sopharak et al. [28] employed fuzzy c-means clustering algorithm to detect hard exudates from low-contrast retinal images. Compared with Un-Machine learning based

methods, Semi-Machine learning based methods can benefit from larger training set by training a model for classification. However, the features are still designed by human in the methods, it being difficult to design features that are most suitable for the applications.

*Full-machine learning based methods:* Both the features used for classification and the classification model are trained over the training set, i.e., the popular deep learning. Tan et al. [14] designed a 10-layer pixel-wise deep convolutional neural network to automatically segment hard exudates, haemorrhages and micro-aneurysms. In this network, the class label of each pixel is predicted by providing a square window, called a patch with size  $51 \times 51$ , centered on that pixel. Prentašić and Lončarić [13] also used a pixel-wise deep convolutional neural network for exudates segmentation. Besides this, they combined the output of bright border probability, vessel probability map and optic disc probability map to produce the final exudate probability map. Apart from pixel-wise deep neural networks for exudates segmentation, Mo et al. [3] designed an image-level fully convolutional residual network (FCRN) to segment exudates. The input to FCRN is a retinal image and it could produce an exudate probability map for only one forward pass. The FCRN shows superior performance than other exudate detection methods. Full-Machine learning based methods can automatically extract the most suitable features and train the model over the training set, fully benefiting from larger training set. Due to the good performance of the deep learning for many applications, we use Full-Machine learning based method in our work.

### 2.2. Class imbalance

For the hard exudates segmentation task, the distribution of hard exudate pixels and background pixels is extremely imbalanced, the hard exudate pixels being much less than background ones in a fundus image. Therefore, the basic sigmoid cross entropy loss is not appropriate as a supervised criteria for hard exudates segmentation, which tends to misclassify the hard exudate pixels as background ones. A common method for addressing this imbalance is to introduce a weight factor  $\beta$  for hard exudate pixels and  $1 - \beta$  for background pixels, i.e., CBCE loss. Xie and Tu [8] used CBCE loss for edge detection. The  $\beta$  is set as the proportion of hard exudate pixels to the whole image. Eigen and Fergus [29] used a frequency-based balance strategy which is similar to class-balanced loss, and their difference lies in the calculation of weight factor. Lin et al. [30] proposed focal loss for dense object detection, which can be taken as a variant of CBCE loss by using adaptive weight for each pixel instead of a constant weight for all pixels in CBCE loss. CBCE loss has been widely applied over many applications in computer vision field due to its good performance in practice. However, CBCE is not appropriate for hard exudates segmentation task due to the following two reasons. First, compared to normal objects such as car, the size of a hard exudate is small, it is difficult to distinguish whether it is a hard exudate or a background noise. Second, the number of hard exudate pixels is much smaller than that of background pixels, so that background pixels with smaller weighted loss are tend to misclassified as hard exudate pixels with larger weighted loss. Therefore, it tends to generate a noisy result that contains many false hard exudates by using CBCE loss, which greatly reduces the performance of hard exudates segmentation.

In practice, the background pixels are much more than the hard exudate pixels, but a large part of background pixels are easy to classify due to the large difference between these background pixels and the hard exudate pixels. Therefore, we designed a top-k loss to focus more over the hard-to-classify background pixels, only the top-k background pixels with high loss being considered in

each iteration of model training. Moreover, we implemented a fast version of top-k loss, named bin loss, that reduces the time complexity from  $O(n \log(n))$  of the top-k loss to  $O(n)$ , where  $n$  is the number of background pixels.

### 3. Method

*Bin Loss* is introduced in detail in this section to address the problem of CBCE loss over background pixels being tend to misclassified as hard exudate pixels. Here we first revisit the cross entropy (CE) loss for binary-class semantic segmentation:

$$L_{CE}(X, Y) = - \sum_{j \in Y_+} \log \Pr(y_j = 1|X; W) - \sum_{j \in Y_-} \log \Pr(y_j = 0|X; W) \quad (1)$$

where  $X$  and  $Y$  denote a retinal fundus image and its ground-truth respectively,  $Y_-$  and  $Y_+$  denote background pixels set and hard exudate pixels set respectively,  $W$  represents model parameters,  $y_j$  represents the label of the pixel  $j$  (0 is background pixel, and 1 is hard exudate pixel),  $\Pr(y_j = q|X; W)$  is the probability of a pixel  $j$  being a hard exudate pixel ( $q = 1$ ) or a background pixel ( $q = 0$ ) outputted by the model.

#### 3.1. CBCE loss function

The distribution of hard exudate pixels and background pixels is highly imbalanced, for example, only 0.91% and 0.13% of pixels are hard exudate ones for IDRiD dataset and e-optha EX dataset, respectively. In this situation, CE loss tends to misclassify hard exudate pixels as background ones, since it can achieve over 99% accuracy if simply classifying all pixels to background ones that is usually the optimal solution for the cases with classes overlapping. CBCE loss is a popular strategy for addressing the class-imbalance by introducing a weight factor for hard exudate pixels and background ones. As described in [8], CBCE loss is defined as:

$$L_{CBCE}(X, Y) = - \beta \sum_{j \in Y_+} \log \Pr(y_j = 1|X; W) - (1 - \beta) \sum_{j \in Y_-} \log \Pr(y_j = 0|X; W) \quad (2)$$

where  $\beta = \frac{|Y_-|}{|Y_-| + |Y_+|}$  is the weight factor, obviously  $\beta \in [0, 1]$ .

#### 3.2. Top-k loss function

CBCE loss uses a weight factor  $\beta$  to balance the importance of hard exudate pixels and background ones for valid training. However, it leads to another problem, i.e., background pixels tend to misclassified as hard exudate ones since the weighted loss for misclassifying a background pixel is much smaller than that for misclassifying a hard exudate pixel. According to Eq. (2), we can get:

$$\frac{\beta}{1 - \beta} = \frac{|Y_-|}{|Y_+|} \quad (3)$$

Therefore, for a hard exudate pixel and a background pixel with the same loss  $L$ , the weighted loss  $WL_h$  for the hard exudate pixel and the weighted loss  $WL_b$  for the background pixel satisfy:

$$WL_h = \frac{|Y_-|}{|Y_+|} WL_b \quad (4)$$

$|Y_+| \ll |Y_-|$ , thus  $WL_h \gg WL_b$ .

As a result, the background pixels, that are similar with hard exudate ones such as vessel reflections, artefacts and soft exudates, tend to be misclassified as the hard exudate ones due to their much lower weighted loss, which causes a noisy result. To address

the loss-unbalance problem, we propose a top-k loss, which considers all hard exudate pixels but only the top-k background pixels with the larger losses:

$$L_{top-k}(X, Y) = - \beta \sum_{j \in Y_+} \log \Pr(y_j = 1|X; W) - (1 - \beta) \sum_{j \in Y_-^{(k)}} \log \Pr(y_j = 0|X; W) \quad (5)$$

where  $Y_-^{(k)}$  denotes the set of the top-k background pixels with the larger loss value, and the weight factor  $\beta = \frac{|Y_-^{(k)}|}{|Y_-^{(k)}| + |Y_+|}$ .

Obviously, the CBCE loss is a special case of the top-k loss. When  $k$  is set as the number of background pixels, the top-k loss degenerates the CBCE loss. Obviously, the same weights can be applied to hard exudate pixels and background ones, if  $k$  is set as the number of hard exudate pixels  $|Y_+|$ , i.e.,  $\beta = \frac{1}{2}$ . Benefitting from ignoring the background pixels that are easy-to-classify, we can focus more over the classification of the background pixels that are hard-to-classify and the hard exudate pixels for improving performance.

#### 3.3. Bin loss function

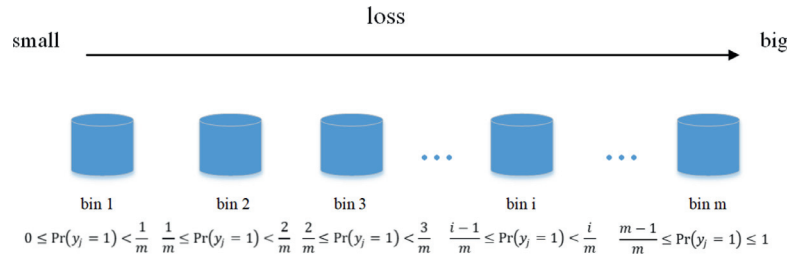
The top-k loss considers both class-unbalance and loss-unbalance, but it is not efficient for practical applications. Assuming a fundus image  $I$  for hard exudates segmentation, the size of  $I$  is  $W * H$ . Since over 99% pixels are background ones, the number of the background pixels  $n$  satisfies  $n > 0.99 * W * H$ . To get the top-k background pixels with the greater loss value, we should sort the  $n$  background pixels by the loss value, and take the first  $k$  background pixels for model training. It is well known that the time complexity of the most fast sorting algorithm is  $O(n \log n)$ , which is time-consuming and greatly increases the model training time.

For efficiency, the bin loss, the fast version of the top-k loss, is implemented. As is shown in Fig. 3, bins set  $B = \{b_1, b_2, \dots, b_m\}$  are generated by evenly splitting  $[0, 1]$ , and the interval of each bin is calculated as:

$$Int(i) = \begin{cases} [\frac{i-1}{m}, \frac{i}{m}) & i < m \\ [\frac{i-1}{m}, 1] & i = m \end{cases} \quad (6)$$

For each background pixel with loss value  $l$ , we put it into the  $i$ th bin if  $l \in Int(i)$ . Obviously,  $i$  and  $l$  satisfy  $i = \min(m, \lfloor ml + 1 \rfloor)$ , thus we can calculate the bin id directly by the loss value. After all background pixels are put into the bins, the background pixels with the larger loss values, used for training the model in current iteration, are generated by collecting the bins with the order of right-to-left, since the losses of background pixels in a right bin are definitely larger than those in a left bin. The collection stops when the number of background pixels in the collected bins are greater or equal to  $k$ . Since the id of the bin that each background pixel being put can be directly calculated according to the loss value, the time complexity of putting  $n$  background pixels into  $m$  bins is  $O(n)$ . And the time complexity of collecting background pixels is also  $O(n)$  due to the number of the collected background pixels is  $\alpha n$ , where  $\alpha$  is a constant that is smaller than 1. Therefore, the time complexity of the bin loss is  $O(n)$ , which is more efficient than  $O(n \log n)$  of the top-k loss. Moreover, the bin loss approximates the top-k loss when  $m \rightarrow \infty$ .

In summary, the bin loss is defined as:



**Fig. 3.** Each background pixel is put into one of  $m$  bins according to its loss. The background pixels in each bin is disordered, but a background pixel in the  $s$ th bin definitely has a larger loss than that in the  $t$ th bin if  $s > t$ . Obviously, for a background pixel  $j$ , the id of the bin that it belongs to can be calculated by  $\min(m, \lfloor m \Pr(y_j = 0|X; W) + 1 \rfloor)$ .

$$L_{bin}(X, Y) = -\beta \sum_{j \in Y_+} \log \Pr(y_j = 1|X; W) - (1 - \beta) \sum_{j \in Y_-^{(B)}} \log \Pr(y_j = 0|X; W) \quad (7)$$

where  $Y_-^{(B)}$  denotes the background pixels collected from the bins that are used for model training, and it is determined by:

$$j^* = \max \left\{ j : \sum_{i=j}^m |b_i| \geq \lambda |Y_+|, j = m, \dots, 2, 1 \right\} \quad (8)$$

where  $Y_+$  denotes the hard exudate pixels set in ground truth  $Y$ , and  $\lambda$  is a hyper-parameter that determines the number of collected background pixels. Obviously, the same weights can be applied to hard exudate pixels and background ones when  $m \rightarrow \infty$ , if  $\lambda$  is set as 1, just like that in the top-k loss.

As a result, the weight factor  $\beta$  of Eq. (7) can be calculated as:

$$\beta = \frac{\sum_{i=j^*}^m |b_i|}{|Y_+| + \sum_{i=j^*}^m |b_i|} \quad (9)$$

## 4. Experiments and results

### 4.1. Semantic segmentation models

We evaluated the bin loss function on three models, including HED, FCRN, and DeepLab v2. DeepLab v2 [15] is a popular semantic segmentation model, and achieved state-of-the-art performance at VOC challenge task. Moreover, FCRN [3] was designed specially for hard exudates segmentation, and achieved the best performance over e-optha EX dataset. Furthermore, HED [8] was first proposed for edge detection. However, we found it behaves well in hard exudates segmentation, especially for small ones. Here we introduce three models in detail.

#### 4.1.1. HED

HED is an image-to-image edge detection network by means of fully convolutional neural network and deeply supervised nets [31]. The basic network of HED for feature extraction is VGGNet [6] with 5 stages. Moreover, HED contains five side-output layers with different receptive fields, and each can be regarded as a stand-alone segmentation model. When it comes to training, there are six loss functions to be optimized in total, in which five corresponds to side-output layer loss, and one corresponds to weight-fusion layer loss. In our experiments, six loss functions are all replaced with bin loss. When it comes to testing, the segmentation result of HED is obtained by fusing five probability maps together through a weight-fusion layer. Benefitting from the weight-fusion layer that fuses multi-scale probability maps, the segmentation result of HED is fine, which is appropriate for the case of small objects segmentation with the requirement of high precision.

#### 4.1.2. FCRN

The FCRN (Fully Convolutional Residual Network) [3] consists of two parts, i.e., downsampling module and upsampling one. The downsampling module is responsible for feature learning, and ResNet-50 [9] is employed for this task. After successive pooling operations, the size of the features map is much smaller than ground truth. Therefore, upsampling operations are applied to get pixel-wise prediction results. In FCRN, skip connections are adopted in upsampling module to incorporate multi-level features so as to get more accurate prediction results, just like what did in [7]. It should be noted that the original FCRN [3] is trained using frequency-based balance strategy [29] which is similar to CBCE loss, while we trained FCRN using CBCE loss or bin loss in our experiments. What's more, compared to the original FCRN [3], we put the whole image as the model input instead of a cropped region that contains hard exudates, so that the results we achieved by the FCRN is incomparable with that in [3].

#### 4.1.3. DeepLab

The DeepLab system [15] once achieved the state-of-the-art performance at semantic image segmentation task of the PASCAL VOC-2012. It uses atrous convolution to effectively enlarge the receptive field to incorporate larger context without increasing the number of parameters or the amount of computation. Moreover, it uses atrous spatial pyramid pooling to segment objects at multi-scales. And, fully connected conditional random field (CRF) is employed to further refine segmentation results, which is not used in our experiments to make the comparison of CBCE loss and bin loss more clear.

### 4.2. Datasets

We evaluated our method over two publicly available datasets, i.e., e-optha EX and IDRiD.

The e-optha EX database [16] is a publicly available dataset and provides pixel level annotation for exudate segmentation. There are 82 retinal images in e-optha EX, of which 47 images with hard exudates, and 35 healthy images. In addition, the resolution of these images range from  $1440 \times 960$  to  $2544 \times 1696$  pixels. In our experiments, we adopted the five-fold cross validation to evaluate the performance of all methods over e-optha EX dataset, since there is no explicit partition of training set and testing set.

The IDRiD (Indian Diabetic Retinopathy Image Dataset) dataset [17] is available at the segmentation and grading of retinal image challenge hosted by 2018 ISBI (International Symposium on Biomedical Imaging) conference. The dataset consists of 81 images that contain hard exudates, each of which has the resolution  $4288 \times 2848$ , and pixel-level annotation is provided. Moreover, explicit partition of training set and testing set are provided in the challenge, 54 images for training and the rest 27 images for testing.

**Table 1**Statistical information of IDRiD and e-optha EX datasets, where  $\text{Ratio} = \frac{\text{hard exudate pixels}}{\text{background pixels}}$ .

| Dataset    | Hard exudate pixels | Background pixels | Ratio | Exudate images | Healthy images | Total images |
|------------|---------------------|-------------------|-------|----------------|----------------|--------------|
| IDRiD      | 890,4004            | 98,028,6140       | 0.91% | 81             | 0              | 81           |
| e-optha EX | 319,626             | 238,800,384       | 0.13% | 47             | 35             | 82           |

The statistical information of these two datasets are shown in Table 1, which shows the distribution of hard exudate pixels and background pixels is highly imbalanced for both IDRiD dataset and e-optha EX dataset, hard exudate pixels being much less than background ones.

#### 4.3. Evaluation metrics

We adopted four metrics to evaluate the segmentation results, including sensitivity (SE), positive predictive value (PPV), F-score, and the area under the SE-PPV curve (AUC). F-score is an important metric, since it is high if and only if both SE and PPV are high. SE, PPV and F-score are defined as:

$$SE = \frac{TP}{TP + FN} \quad (10)$$

$$PPV = \frac{TP}{TP + FP} \quad (11)$$

$$F\text{-score} = \frac{2 \times SE \times PPV}{SE + PPV} \quad (12)$$

The number of true positives (TP), false positives (FP) and false negatives (FN) can be calculated by directly counting the number of correctly classified pixels. However, for hard exudates segmentation task, the result is acceptable if the overlap ratio of a detected hard exudate and the ground truth is larger than a threshold. Therefore, we used the connected component level validation proposed by Zhang et al. [32] to calculate TP, FN and FP. The segmentation results for each image can be divided into a set of candidates  $P = \{P_1, P_2, \dots, P_M\}$ , where  $P_i$  corresponds to a connected component. Also, ground truth for each image can be divided into a set of candidates  $G = \{G_1, G_2, \dots, G_N\}$ .

A pixel is considered to be true positive if, and only if, it belongs to any of the following sets:

- $P_i \cap G_j$ ;
- $P_i$  that satisfies  $\frac{|P_i \cap G|}{|P_i|} > \sigma$ ;
- $G_j$  that satisfies  $\frac{|G_j \cap P|}{|G_j|} > \sigma$ .

where  $|\cdot|$  is the number of elements in set, and  $\sigma \in [0, 1]$  is a parameter that controls the overlap ratio threshold.

A pixel is considered as false positive if, and only if, it belongs to any of the following sets:

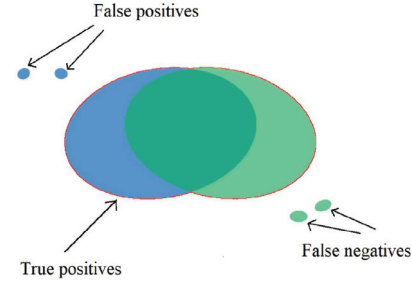
- $P_i$  that satisfies  $P_i \cap G = \emptyset$ ;
- $P_i \cap \bar{G}$  that satisfies  $\frac{|P_i \cap \bar{G}|}{|P_i|} \leq \sigma$ , where  $\bar{G}$  is the complement to  $G$ .

A pixel is considered as false negative if, and only if, it belongs to any of the following sets:

- $G_j$  that satisfies  $G_j \cap P = \emptyset$ ;
- $G_j \cap \bar{P}$  that satisfies  $\frac{|G_j \cap \bar{P}|}{|G_j|} \leq \sigma$ , where  $\bar{P}$  is the complement to  $P$ .

All other pixels are considered as true negatives. Fig. 4 shows the definition of true positives, false positives and false negatives for component level validation.

As for the segmentation task, the output of the deep neural network is a probability map, and the calculation of TP, FP and FN is highly dependent on a threshold used for binarizing the map. The pixel with probability less than the threshold is classified as the



**Fig. 4.** Illustration of the definition of true positives, false positives and false negatives for component level validation. The blue regions represent the detected hard exudates, and the green regions represent ground truth.

background one, and the pixel with probability greater than the threshold is classified as the hard exudate one. Therefore, only SE and PPV are not enough to evaluate the segmentation result, so that a SE-PPV curve is given by varying the threshold from 0 to 1. The area under the SE-PPV curve (AUC) is employed as the most important indicator for evaluating segmentation result in our case, which ranges from 0 to 1. The better the segmentation result is, the larger the AUC is.

#### 4.4. Implementation details

##### 4.4.1. Model implementation

The proposed bin loss function and the segmentation networks were implemented using a publicly available convolutional network framework – Caffe [33].

##### 4.4.2. Data preprocessing and augmentation

Since the resolution of training images is too large to fit them into GPU memory, we resized each image of two datasets to  $1440 \times 960$  (width  $\times$  height) before training. And the corresponding ground truth were also resized to  $1440 \times 960$ .

To improve the performance of models, data augmentation was applied to generate more training samples. The factor for data augmentation was set as 5. And we used various transformations to augment the training set, including rotation by an angle of 90, 180 and 270, and flipping horizontally and vertically.

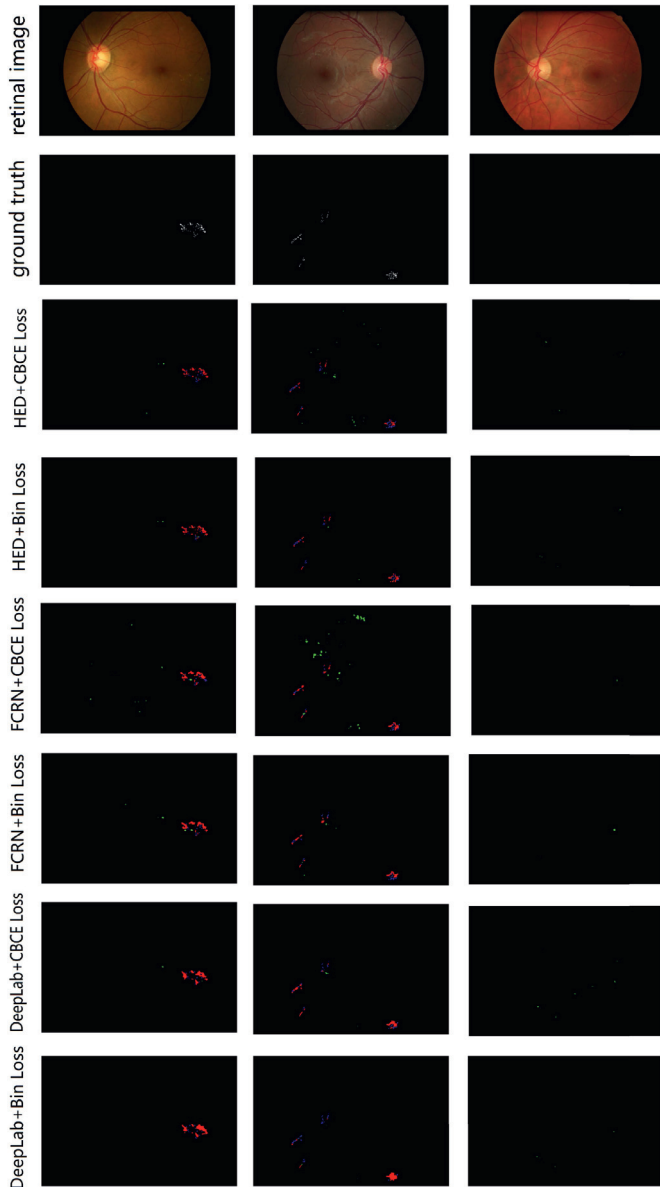
##### 4.4.3. Parameter settings

The settings of hyper-parameters for HED, FCN and DeepLab are summarized in Table 2. What's more, in the training phase, retinal images are cropped randomly in DeepLab. However, in the testing phase, inputs to DeepLab is not cropped so that we get a hard exudates probability map for the whole image. One thing worth noting is that we did not use CRF in our experiments, since the output of CRF is a binary map instead of a probability map, which is not appropriate for the computation of evaluation metrics. The deconvolutional layers in all three segmentation models are initialized by bilinear interpolation. At last, the hyper-parameters  $\lambda$  and  $m$  in Eq. (8) are set to 20 and 5, respectively.

After training, all three segmentation networks take an image with size  $1440 \times 960$  as input, and produce a corresponding hard exudates probability map of the same size as the output. One thing worth noting is the probability maps are stored in matlab format of official DeepLab system and they cannot be converted to images

**Table 2**  
Parameter settings for HED, FCRN and DeepLab.

| Model   | Initial learning rate | Learning rate policy | Momentum | Weight decay | Batch size | Iterations |
|---------|-----------------------|----------------------|----------|--------------|------------|------------|
| HED     | 1e-8                  | Step                 | 0.9      | 0.0005       | 1          | 20,000     |
| FCRN    | 1e-6                  | Step                 | 0.9      | 0.0005       | 10         | 1500       |
| DeepLab | 1e-6                  | Poly                 | 0.9      | 0.002        | 1          | 200,000    |

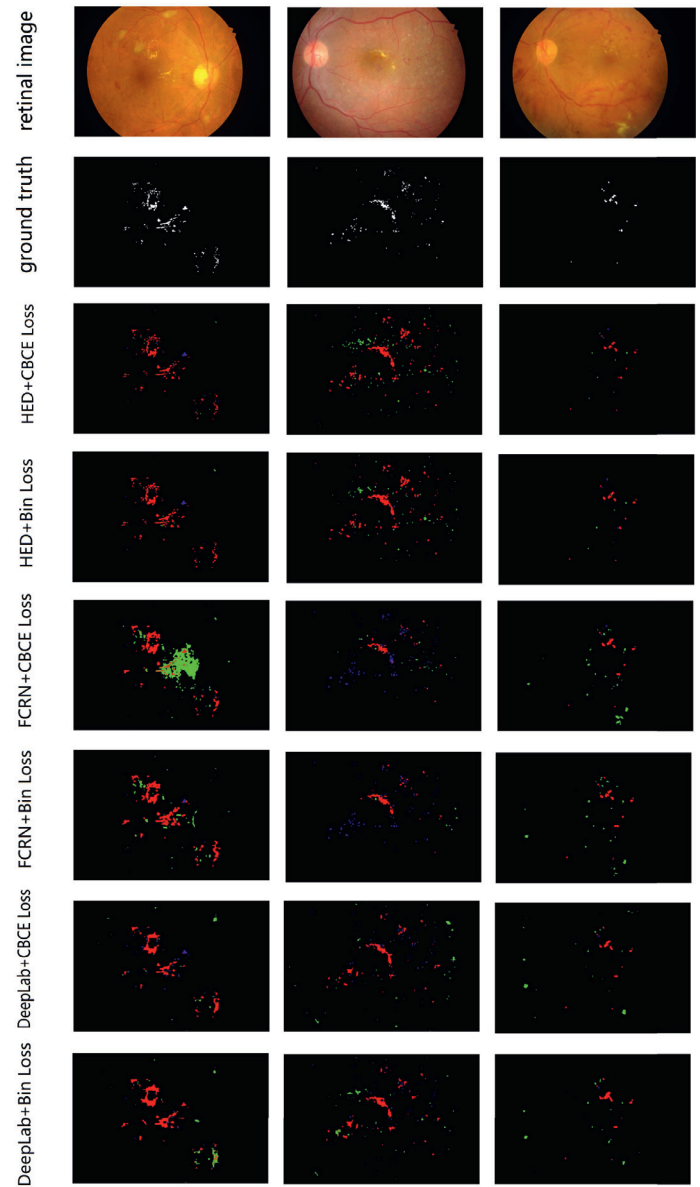


**Fig. 5.** Comparison of binary segmentation results of CBCE Loss and Bin Loss over e-optha EX dataset (red regions denote true positives, green regions denote false positives, and blue regions denote false negatives).

directly, extra flipping and rotating operations are needed. When it comes to calculating the evaluation metrics, the probability map is resized back to its original size so that it is the same size with ground truth. According to Zhang et al. [32], we set overlap ratio  $\sigma = 0.2$  to calculate SE and PPV in our experiments.

#### 4.4.4. Running environment

The experimental platform is a workstation equipped with two Intel E5 processors each with six 2.1 GHz cores, 128GB memory and two NVIDIA GTX 1080ti GPUs, and the operating system is



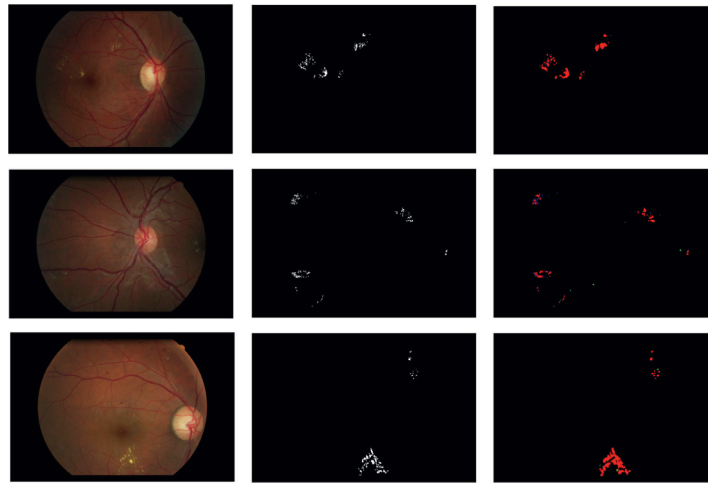
**Fig. 6.** Comparison of binary segmentation results of CBCE Loss and Bin Loss over IDRiD dataset (red regions denote true positives, green regions denote false positives, and blue regions denote false negatives).

Ubuntu 16.04. In addition, we use MATLAB R2014a for data augmentation and the calculation of evaluation metrics.

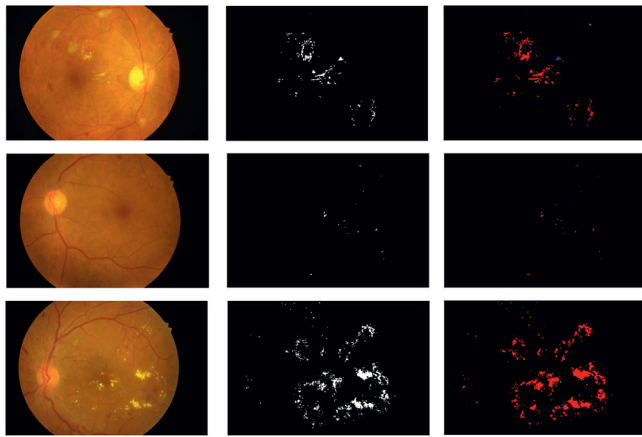
#### 4.5. Results

##### 4.5.1. Visualization of segmentation results

For each segmentation model, the output is a probability map for hard exudates, and each value of the map represents the probability of each pixel that belongs to exudate class. We show the



**Fig. 7.** Binary segmentation results of HED trained with bin loss using component level validation over e-optha EX dataset. Columns from left to right are the retinal image, ground-truth and binary segmentation results (red regions denote true positives, green regions denote false positives, and blue regions denote false negatives), respectively.



**Fig. 8.** Binary segmentation results of HED trained with bin loss using component level validation over IDRiD dataset. Columns from left to right are the retinal image, ground-truth and binary segmentation results (red regions denote true positives, green regions denote false positives, and blue regions denote false negatives), respectively.

segmentation results of CBCE loss and bin loss on three models in Fig. 5 and Fig. 6, respectively.

We can observe from the segmentation results of the first and second retinal images of Fig. 5 that HED and FCNR trained with CBCE loss are easy to misclassify some similar structures (vessel reflections in Fig. 2(c), and soft exudates in Fig. 2(d)) as hard exudate pixels. While for the results with bin loss, the hard-to-classify background pixels are given lower probabilities in the probability map, and the misclassified background pixels has been obviously reduced. In addition, the segmentation results of the first retinal images of Fig. 6 shows that FCNR trained with CBCE loss misclassifies vessel reflections and part of optic disc as hard exudates. And the third retinal image of Fig. 6 indicates that FCNR gives high probabilities to soft exudate regions. However, when FCNR is trained using bin loss, soft exudate regions are given much lower probabilities and a large part of the misclassification has been improved. These observations over e-optha EX and IDRiD datasets indicate the bin loss shows superior performance on hard negative pixels over the CBCE loss.

Besides the visualization of probability maps, the component level evaluation results of HED trained using bin loss over e-optha EX and IDRiD datasets are shown in Figs. 7 and 8. From the first

**Table 3**

Experimental results over e-optha EX dataset (best results are shown in bold).

| Model                | SE            | PPV           | F-score       | AUC           |
|----------------------|---------------|---------------|---------------|---------------|
| HED+CBCE Loss        | 0.8334        | 0.8029        | 0.8179        | 0.8750        |
| HED+Bin Loss         | <b>0.8644</b> | <b>0.8759</b> | <b>0.8701</b> | <b>0.9184</b> |
| FCNR+CBCE Loss       | <b>0.7995</b> | 0.7849        | 0.7921        | 0.8120        |
| FCNR+Bin Loss        | 0.7835        | <b>0.8152</b> | <b>0.7990</b> | <b>0.8257</b> |
| DeepLab v2+CBCE Loss | 0.8384        | 0.7641        | 0.7995        | 0.7926        |
| DeepLab v2+Bin Loss  | <b>0.8472</b> | <b>0.8577</b> | <b>0.8524</b> | <b>0.8992</b> |

row of Fig. 7, we can observe that there exists vessel reflections in retinal image, but the binary segmentation map is robust to this kind of noise. In addition, the second retinal image of Fig. 7 contains artefacts around optic disc and vessels, however, the binary segmentation result is satisfactory. What's more, from the first row of Fig. 8 we can observe that there exists vessel reflections and many soft exudates in retinal image. Furthermore, in the second row of Fig. 8, the hard exudate pixels are largely unnoticeable, unlike the first and the third images. Although these problems exists, the corresponding binary segmentation maps are satisfied. All these examples over e-optha EX and IDRiD datasets illustrate the effectiveness of bin loss in hard negatives classification.

#### 4.5.2. Statistical results

The statistical results of SE, PPV, F-score and AUC are described here. To plot the SE-PPV curve, SE and PPV is calculated with different 256 thresholds, so that 256 pairs of SE and PPV are obtained. Then, the SE-PPV curve is plotted with the 256 pairs of points. The SE-PPV curves over e-optha EX and IDRiD datasets are shown in Figs. 9 and 10, respectively.

To calculate SE and PPV, a threshold is needed for binarizing the probability map of hard exudates. We select the optimal operation point of SE-PPV curve as threshold. Then, the statistical results of CBCE loss and bin loss over e-optha EX and IDRiD are shown in Tables 3 and 4, respectively.

We can observe from Table 3 that AUC was increased by more than 10 percentage points, and PPV was increased by more than 9.3 percentage points over DeepLab by replacing the CBCE loss with the bin loss. Even for FCNR, AUC was increased by more than 1.3 percentage points, and PPV was increased by more than 3 percentage points. In addition, all of the four evaluation metrics are improved on HED.

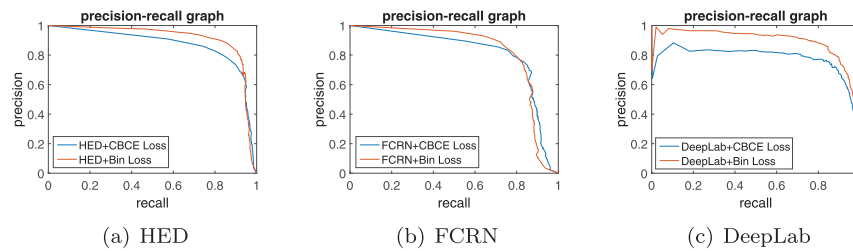


Fig. 9. The SE-PPV curves for HED, FCNR and DeepLab over e-optha EX dataset.

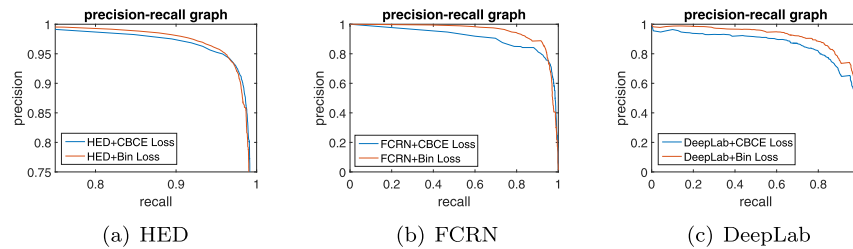


Fig. 10. The SE-PPV curves for HED, FCNR and DeepLab over IDRid dataset.

Table 4

Experimental results over IDRid dataset (best results are shown in bold).

| Model                | SE            | PPV           | F-score       | AUC           |
|----------------------|---------------|---------------|---------------|---------------|
| HED+CBCE Loss        | <b>0.9579</b> | 0.9497        | 0.9538        | 0.9855        |
| HED+Bin Loss         | 0.9574        | <b>0.9540</b> | <b>0.9557</b> | <b>0.9871</b> |
| FCNR+CBCE Loss       | 0.8819        | 0.8421        | 0.8616        | 0.9085        |
| FCNR+Bin Loss        | <b>0.9167</b> | <b>0.8884</b> | <b>0.9023</b> | <b>0.9485</b> |
| DeepLab v2+CBCE Loss | 0.7968        | 0.8219        | 0.8092        | 0.8639        |
| DeepLab v2+Bin Loss  | <b>0.8448</b> | <b>0.8542</b> | <b>0.8495</b> | <b>0.9162</b> |

Table 5

Running time comparison over CBCE loss, top-k loss and bin loss. The unit of time is milliseconds.

| Time     | CBCE loss | Top-k loss | Bin loss |
|----------|-----------|------------|----------|
| Forward  | 258.448   | 326.99     | 261.588  |
| Backward | 9.809     | 337.27     | 18.677   |

Moreover, We can observe from Table 4 that bin loss leads to a good promotion on three models. On one hand, the AUC, F-score and PPV are all increased by at least 4 percentage points on FCNR. On the other hand, the AUC, F-score and SE are increased by more than 4 percentage points on DeepLab.

It can be seen from Tables 3 and 4 that PPV score has increased on all experiments, which means fewer pixels are classified as false positives, and it is consistent with the visualization results of the probability maps in Fig. 5. All these statistical results indicate that there is a significant improvement on all three models by replacing the CBCE loss with the bin loss.

#### 4.5.3. Running time comparison

We measure the running time of both forward and backward over CBCE loss, top-k loss and bin loss using tools provided by Caffe, and the average running time over 100 iterations are shown in Table 5. We can observe that the running speed of bin loss is very close to CBCE loss, and it is much faster than top-k loss.

#### 4.6. Parameter selection

For the worst cases, if all background pixels are selected to be optimized in bin loss, then it degenerates to CBCE loss. As a result, bin loss does not take any effect. Here we provide several

principles for selecting  $m$  and  $\lambda$ . Bigger  $m$  corresponds to fine-grained partition of background points and longer running time, and smaller  $m$  means coarse-grained partitioning of background points and shorter running time. When  $m \rightarrow \infty$ , bin loss approximates the top-k loss. We suggest  $m$  falls into  $[2, 100]$ . In addition, the setting of  $\lambda$  is related to the ratio of positive pixels over an image, and it must enable at least one bin instead of all bins to be selected.

## 5. Conclusion

In this paper, we proposed the top-k loss and its fast implementation named bin loss for hard exudates segmentation. The bin loss considers a part of background pixels that with larger loss values and all hard exudate pixels for training, which avoids the loss-unbalance problem in the CBCE loss by focusing more over hard-to-classify background pixels. We evaluated the bin loss on three popular semantic segmentation models, HED, DeepLab v2 and FCNR, over two publicly available datasets, e-optha EX and IDRid. Segmentation results indicate that compared with the CBCE loss, bin loss is more robust to hard-to-classify background structures such as soft exudate, vessel reflections and artefacts. In addition, statistical results show that bin loss leads to higher PPV, F-score and AUC on all experiments. Extensive experiments demonstrate that the proposed bin loss achieves superior performance than the CBCE loss on the hard exudates segmentation task. However, there still exists misclassification problems using bin loss, and the solutions to this problem are one of our future research directions.

## Conflict of interest

None.

## Acknowledgments

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